

Grid Distillation: Compositional Image Distillation via Structured Generative Grids

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Paper



Video



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Motivation

Deep learning demands millions of training images — creating enormous costs in storage, compute, and energy, with growing concerns about privacy and reproducibility.

Dataset Distillation aims to synthesize a small set of images that trains a model nearly as well as the full dataset — compressing an entire class into just 10–50 representative samples.

The core challenge: how do we maximally preserve semantic diversity, world knowledge, and spatial structure in a handful of images?



Interpretability: Low	Interpretability: High	Interpretability: High	Interpretability: High
Information Density: Moderate	Information Density: Moderate	Information Density: Low	Information Density: Very High
World Knowledge: No	World Knowledge: No	World Knowledge: Yes	World Knowledge: Yes

Prior Art

Patch-based methods (RDED):

- Stitch random crops — discard global layout, No injected world knowledge, Limited intra-class diversity

Diffusion-based methods (VLCP, Minimax):

- Generate each image independently, Rich priors but no inter-instance context, Spatially shallow — no compositional structure

Gap: No method jointly captures compositional structure + world knowledge.

Contributions

- SSDIM:** Spectral submodular selection unifying coverage, diversity & manifold fidelity.

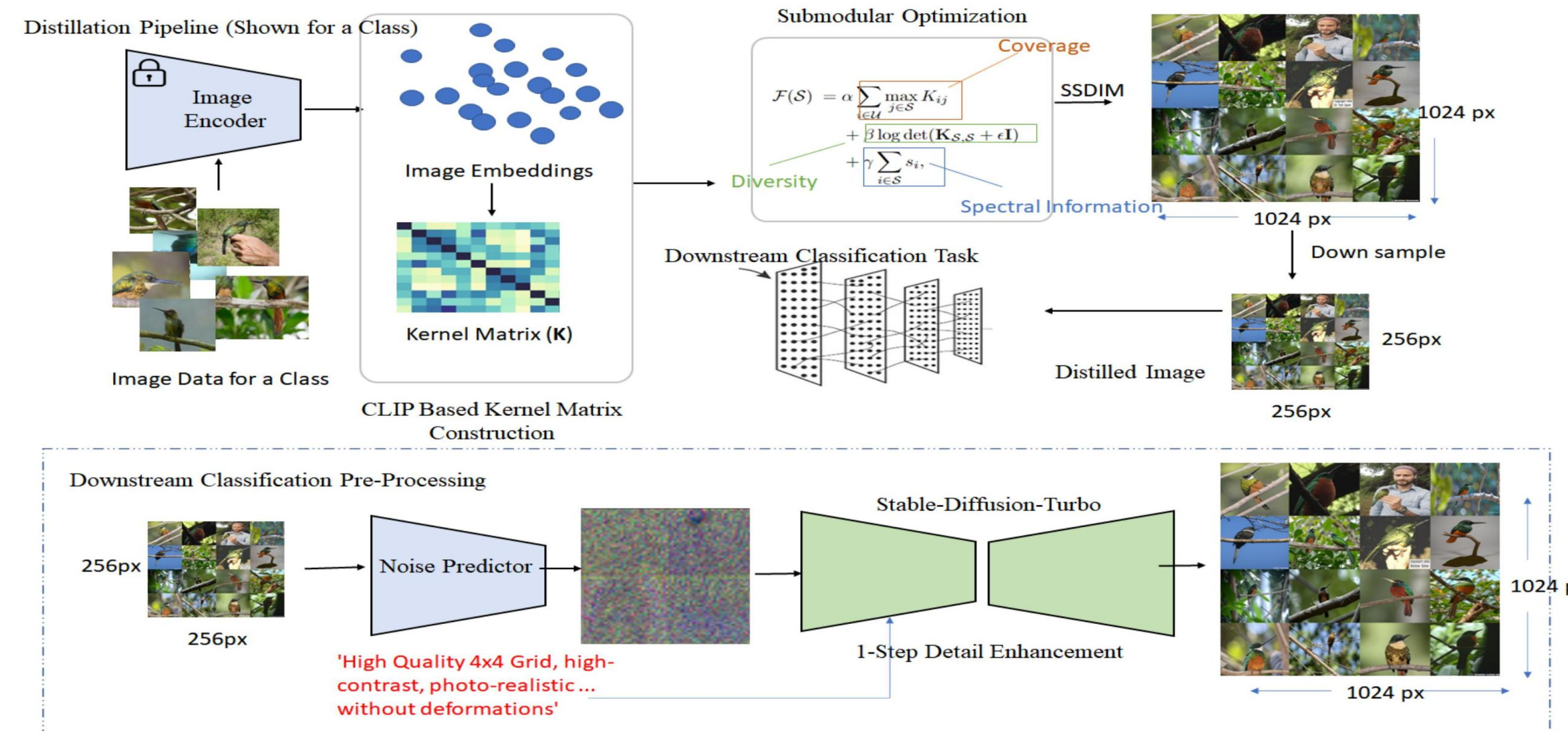
- 1-step diffusion** restoration via InvSR — only +16.9% one-time overhead.

- Grid-aware cropping** preserves compositional cues at train time; drop-in at test time.



Random selection → redundant grids SSDIM → semantically diverse grids

Method Overview



SSDIM Selection

Select L^2 exemplars maximizing coverage, diversity, and spectral informativeness:

$$\mathcal{F}(\mathcal{S}) = \alpha \sum_{i \in \mathcal{U}} \max_{j \in \mathcal{S}} K_{ij} + \beta \log \det(\mathbf{K}_{\mathcal{S}, \mathcal{S}} + \epsilon \mathbf{I}) + \gamma \sum_{i \in \mathcal{S}} s_i$$

$\alpha=1.0$ (Coverage) $\beta=0.6$ (Diversity) $\gamma=0.3$ (Spectral)

Solved via Frank-Wolfe relaxation → top- L^2 rounding → greedy swap refinement.

Runtime: ~9–70s per class on A6000 GPU.

Algorithm 1 — SSDIM

```

Input:  $\mathbf{K}$ ,  $\mathbf{s}$ ,  $L$ ,  $\alpha$ ,  $\beta$ ,  $\gamma$ 
Output:  $\mathcal{S}$ ,  $|\mathcal{S}| = L^2$ 

Init:  $\mathbf{p}$  uniform
        $\mathbf{S} = \emptyset$ 

for  $t = 1$  to  $T$ :
    // Frank-Wolfe
     $\nabla \mathcal{F} = \alpha \mathbf{K} \mathbf{p}$ 
     $\mathbf{v} = \arg \max(\mathbf{e}_i, \nabla \mathcal{F}(\mathbf{p}))$ 
     $\mathbf{p} \leftarrow (1 - \eta_t) \mathbf{p} + \eta_t \mathbf{v}$ 
     $\eta_t = 2 / (t + 2)$ 

    // Round to  $\mathbf{S}$ 
     $\mathbf{S} \leftarrow \text{top-}L^2 \text{ by largest } \mathbf{p}_i$ 

    // Greedy refine
    repeat:
         $\Delta = \mathcal{F}(\mathbf{S} \setminus \{y\} \cup \{x\}) - \mathcal{F}(\mathbf{S})$ 
        if  $\Delta > 0$ : swap
    until no gain

return  $\mathbf{S}$ 

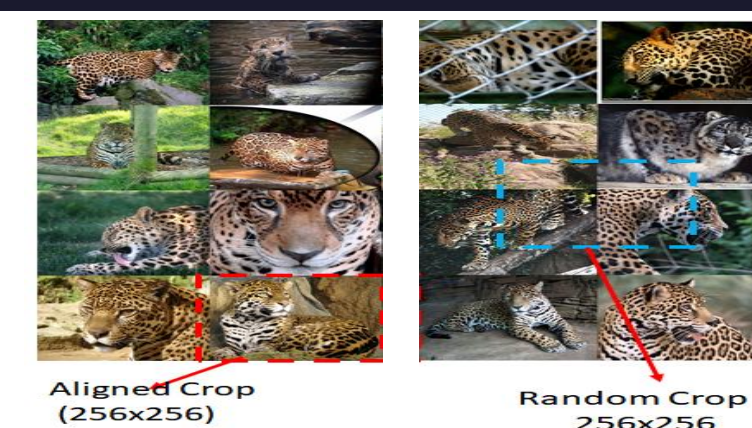
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Grid Aware Cropping

Mix two crop strategies during training:

- p=0.6** → Aligned crop at grid boundaries
- p=0.4** → Random crop for local variation

Test time: standard 224×224 — no changes.



Diffusion Detail Enhancement

Down sampling loses fine-grained spatial detail. We adapt InvSR: 1-step diffusion restores high-frequency textures via CLIP-conditioned class-aware prompts. Only +16.9% overhead.

Dataset	Bilinear Upsampling			Detail Enhancement (Ours)		
	IPC 10	IPC 20	IPC 50	IPC 10	IPC 20	IPC 50
ImageDC	74.7±0.2	79.8±1.3	87.7±0.5	73.5±1.5	79.7±0.7	87.1±0.5
ImageNette	81.5±0.9	85.7±1.0	91.5±0.5	83.3±0.6	87.3±0.5	92.7±0.7
ImageWoof	62.5±0.7	73.1±0.3	82.2±0.7	63.9±0.6	73.7±0.7	82.4±0.2

On ImageNet-1K (IPC=10): Bilinear → 35.4% vs Diffusion Enhancement → **50.0% (+14.6pp)**

Results — ImageWoof

Comparison against 10 dataset distillation methods on ImageWoof (10 fine-grained dog breeds)
10 baselines · 3 architectures · IPC 10→100. Ours consistently leads — **gap widens at low IPC.**

IPC (Ratio)	Test Model	Random	K-Center	Herding	DiT	DM	IDC-1	GLaD	Minimax	D ⁴ M	VLCP	Ours
10 (0.8%)	ConvNet-6	24.3±1.1	19.4±0.9	26.7±0.5	34.2±1.1	26.9±1.2	33.3±1.1	33.8±0.9	33.3±1.7	29.4±0.9	34.8±2.4	49.7±0.7
	ResNetAP-10	29.4±0.8	22.1±0.1	32.0±0.3	34.7±0.5	30.3±1.2	39.1±0.5	32.9±0.9	36.2±3.2	33.2±2.1	39.5±1.5	63.9±0.6
	ResNet-18	27.7±0.9	21.1±0.4	30.2±1.2	34.7±0.4	33.4±0.7	37.3±0.2	31.7±0.8	35.7±1.6	32.3±1.2	39.9±2.6	65.5±0.8
50 (3.8%)	ConvNet-6	41.3±0.6	36.5±1.0	40.3±0.7	46.5±0.8	44.4±1.0	43.9±1.2	-	50.9±0.8	47.4±0.9	54.5±0.6	79.0±0.6
	ResNetAP-10	47.2±1.3	40.6±0.4	49.1±0.7	49.3±0.2	47.1±1.1	48.3±1.0	-	53.9±0.7	51.7±3.2	57.3±0.5	82.4±0.2
	ResNet-18	47.9±1.8	39.6±1.0	48.3±1.2	50.1±0.5	46.2±0.6	48.3±0.8	-	53.7±0.6	53.7±2.2	58.9±1.5	84.3±0.9
100 (7.7%)	ConvNet-6	52.2±0.4	45.1±0.5	54.4±1.1	53.4±0.3	55.0±1.3	53.2±0.9	-	57.8±0.9	57.9±1.5	62.7±1.4	80.7±0.6
	ResNetAP-10	59.4±1.0	54.8±0.2	61.7±0.9	58.3±0.8	56.4±0.8	56.1±0.9	-	62.7±1.4	59.5±1.8	65.7±0.5	83.4±0.7
	ResNet-18	61.5±1.3	50.4±0.4	59.3±0.7	58.9±1.3	60.2±1.0	58.3±1.2	-	62.7±0.4	63.8±1.3	68.3±0.4	85.9±0.2

Results — Cross Dataset Generalization

ImageNette · ResNetAP-10 · IPC 10/20/50

Method	IPC=10	IPC=20	IPC=50
Random	54.2±1.6	63.5±0.5	76.1±1.1
DiT	59.1±0.7	64.8±1.2	73.3±0.9
DM	60.8±0.6	66.5±1.1	76.2±0.4
Minimax	57.7±1.2	64.7±0.8	73.9±0.3
D ⁴ M	60.9±1.7	66.3±1.3	77.7±1.1
VLCP	64.8±3.6	71.4±0.5	81.2±0.8
Grid-Distil (Ours)	83.3±0.6	87.3±0.5	92.7±0.7

ImageNet-1K · IPC=10 · ResNet-18

Method	Venue	Mean	Std
SRe ² L	NeurIPS'23	21.3	0.6
RDED	CVPR'24	42.0	0.1
DiT	CVPR'24	39.6	0.4
Minimax	CVPR'24	44.3	0.5
VLCP	ICCV'25	46.7	0.4
Ours (Bilinear)	—	35.40	0.25
Ours (Detail Enhancement)	—	50.01	0.29

Ablation: SSDIM Components

Isolating each component on real ImageWoof grids
(IPC=20, ResNetAP-10):

- Coverage (α)** alone → 78.2% Most important component
- Diversity (β)** alone → 74.4% Helps but not dominant
- Spectral (γ)** alone → 75.8% Captures manifold structure
- Best config:** $\alpha=1.0$, $\beta=0.6$, $\gamma=0.3$ → 78.6% (α -dominant)

Random baseline: 74.6%

ID	α	β	γ	Top1 Acc	Description
-	0.0	0.0	0.0	74.6	Random
1	0.0	0.0	1.0	75.8	Only γ active
2	0.0	1.0	0.0	74.4	Only β active
3	0.3	0.6	1.0	76.0	γ -dominant
4	0.3	1.0	0.6	74.6	β -dominant
5	0.6	0.3	1.0	73.0	γ -dominant
6	0.6	0.6	0.6	75.6	Balanced regime
7	0.6	1.0	0.3	77.8	β -dominant
8	1.0	0.0	0.0	78.2	Only α active
9	1.0	0.3	0.6	78.0	α -dominant
10	1.0	0.6	0.3	78.6	α -dominant